

A METHOD FOR DERIVING THE ASYMPTOTIC DISTRIBUTION OF ONE-SIDED STOPPING RULES

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SUMMARY

In this paper we give a theorem from which the asymptotic distribution of one-sided stopping rules may be obtained. Siegmund (1968) and Bhattacharya and Mallik (1973) derived the asymptotic distributions in certain cases, but their methods require that the stopping rules be defined in terms of partial sums. Our method does not have this requirement and consequently our limiting distributions are not necessarily normal. Furthermore, our method covers the examples discussed by the above-mentioned authors and also certain examples which cannot be solved by their methods. For instance, the asymptotic distribution of the stopping rule for the estimation of a binomial proportion θ for the case $\theta = \frac{1}{2}$ is derived in Section 2. (In Theorem 3 of Bhattacharya and Mallik (1973) only asymptotic bounds for this distribution are given.) Also the asymptotic distribution of a stopping rule based on sample minima follows in a direct way from our theorem. In this case the asymptotic distribution is exponential.

In Section 1 the main theorem is stated and proved, while in Section 2 applications are considered.

1. Main theorem

The following lemma will be needed in the proof of the theorem:

Lemma:

Let $\{X_n\}$, $\{Y_n\}$, $\{Y'_n\}$ be sequences of random variables such that $Y_n = o_p(1)$, $Y'_n = o_p(1)$ as $n \rightarrow \infty$. Let G be a distribution function. If, for each point of continuity x of G

$$\liminf_{n \rightarrow \infty} P(X_n + Y_n < x) \geq G(x)$$

and $\limsup_{n \rightarrow \infty} P(X_n + Y'_n < x) \leq G(x)$

then $X_n \xrightarrow{D} G$

Proof:

Let $\epsilon > 0$ be arbitrary.

$$\begin{aligned} P(X_n + Y_n < x) &= P(X_n + Y_n < x, Y'_n - \epsilon < Y_n) + P(X_n + Y_n < x, Y'_n - \epsilon \geq Y_n) \\ &\leq P(X_n + Y'_n < x + \epsilon) + P(Y'_n - Y_n \geq \epsilon) \end{aligned}$$

so that, from the assumptions,

$$\begin{aligned} \limsup_{n \rightarrow \infty} P(X_n + Y_n < x) &\leq \limsup_{n \rightarrow \infty} P(X_n + Y'_n < x + \epsilon) \\ &\leq G(x + \epsilon) \end{aligned}$$

Now let $\epsilon \rightarrow 0$, to obtain

$$\limsup_{n \rightarrow \infty} P(X_n + Y_n < x) \leq G(x)$$

This, together with the first assumed inequality, implies $X_n + Y_n \xrightarrow{D} G$ and the lemma follows, since $Y_n \xrightarrow{P} 0$.

Theorem:

Define the stopping rule N as the first integer $n \geq n_0$ (where $n_0 \geq 1$) such that $T_n \geq c_n$, where $\{T_n\}$ is a sequence of statistics and where $\{c_n\}$ is a sequence of constants, depending on a parameter α . Assume that for some function g and constant θ

$$(1.1) \quad g(n)/(T_n - \theta) \xrightarrow{D} G \text{ as } n \rightarrow \infty$$

where G is a continuous distribution function, and for some non-negative function d (which may be a random variable)

$$(1.2) \quad d(\alpha)/(T_N - \theta) \xrightarrow{D} G \text{ as } \alpha \rightarrow 0$$

Let $h(n, \alpha)$ be a function defined for each integer n and real $\alpha > 0$, such that h is strictly decreasing in n for each fixed α . For simplicity we write $h(n)$ instead of $h(n, \alpha)$. Let h satisfy

$$(1.3) \quad h(m) \rightarrow x \text{ as } \alpha \rightarrow 0$$

where, for any real x fixed, $m = m(\alpha, x)$ is defined as a function of α and x by

$$(1.4) \quad g(m)/(c_m - \theta) = x,$$

and satisfying $m(\alpha, x) \rightarrow \infty$ as $\alpha \rightarrow 0$ for each x . Also, $h(\cdot)$ must satisfy the requirement

$$(1.5) \quad d(\alpha)(c_N - \theta) - h(N) = o_p(1) \text{ as } \alpha \rightarrow 0.$$

$$\text{Then } h(N) \xrightarrow{D} G \text{ as } \alpha \rightarrow 0.$$

Under the above conditions, the same conclusion holds if N is defined as the first integer $n \geq n_0$ such that $T_n \leq c_n$, provided h is increasing.

Note: In the applications considered in Section 2 one arrives at (1.2) by first proving that $N/b(\alpha) \rightarrow 1$ a.s. as $\alpha \rightarrow 0$ for some function $b(\alpha) \rightarrow \infty$ as $\alpha \rightarrow 0$. From this and (1.1) together with results of Anscombe (1952), (1.2) then follows. The function $d(\alpha)$ is either $g(N)$ or the asymptotic expression for $g(N)$, while the function $h(N)$ is of the form

$$k_1(\alpha)(N - b(\alpha)) + k_2(\alpha)$$

with k_1 and k_2 obtained from (1.3). To verify (1.5), techniques similar to those used by Gut (1974) may be used.

Proof of the theorem:

Let m satisfy (1.4), then

$$\begin{aligned} & P(h(N) \geq h(m)) \\ &= P(N \leq m) \\ &\geq P(T_m \geq c_m) \text{ by definition of } N \\ &= P(g(m)(T_m - \theta) \geq g(m)(c_m - \theta)) \\ &= P(g(m)(T_m - \theta) \geq x) \end{aligned}$$

Hence, using (1.3)

$$\limsup_{\alpha \rightarrow 0} P(h(N) + o(1) < x) \leq G(x)$$

To obtain the converse inequality, note that the definition of N and (1.2) imply

$$\begin{aligned} & \liminf_{\alpha \rightarrow 0} P(d(\alpha)(c_N - \theta) < x) \\ &\geq \liminf_{\alpha \rightarrow 0} P(d(\alpha)(T_N - \theta) < x) \\ &= G(x) \end{aligned}$$

This, together with (1.5), shows that

$$\liminf_{\alpha \rightarrow 0} P(h(N) + o_p(1) < x) \geq G(x)$$

The theorem now follows on an application of the lemma.

Remark:

In the first part of the proof we tacitly assumed that the m defined by (1.4) is an integer. The problem may be circumvented by using in the proof $[m]$, the largest integer contained in m . This may be done if, for instance, $g(m)(c_m - c_{[m]}) \rightarrow 0$ and $g([m])/g(m) \rightarrow 1$ as $\alpha \rightarrow 0$. These conditions are satisfied in all the applications considered.

2. Applications

- (1) The following stopping rule was introduced by Swanepoel and Geertsema (1976) in a discussion of tests of power one for the location parameter of an exponential distribution: Let $X_1, X_2, \dots$ be independent random variables with common density $\exp\{-(x - \theta)\}$, $x \geq \theta$, $M_n = \min(X_1, \dots, X_n)$, $c_n = (n - 2)^{-1} \log(1 + (n - 2)(2\alpha)^{-1})$. The stopping rule N is the first $n \geq 3$ such that $M_n \geq c_n$.

If $h(N) = -\theta(N - \theta^{-1} \log \alpha^{-1}) + \log \log \alpha^{-1} + 2\theta + \log \theta^{-1} - \log 2$ we show that

for $\theta > 0$ $h(N) \xrightarrow{D} X$ as $\alpha \rightarrow 0$ where X has density $\exp\{-x\}$, $x > 0$.

First note that $N/\lambda \rightarrow 1$ a.s. as $\alpha \rightarrow 0$, where $\lambda = \theta^{-1} \log \alpha^{-1}$ and that $n(M_n - \theta) \xrightarrow{D} X$ as $n \rightarrow \infty$. From these and Anscombe (1952) follows $N(M_N - \theta) \xrightarrow{D} X$ as $\alpha \rightarrow 0$. Here $g(n) = n$, $d(\alpha) = N$. From (1.4) follows $m/\lambda \rightarrow 1$ as $\alpha \rightarrow 0$ so that $h(m) \rightarrow x$. In this example (1.5) follows in a similar way.

- (2) Bhattacharya and Mallik (1973) consider the following stopping rule (their Theorem 3): Let $X_1, X_2, \dots$ be independent random variables taking values 0 and 1 with probabilities $1 - \theta$ and θ respectively, $0 < \theta < 1$, $S_n = X_1 + \dots + X_n$ and N is the first n for which $(S_n + \frac{1}{2})(n - S_n + \frac{1}{2}) \leq cn(n + 1)^3$.

$$\text{i.e. } V_n \leq c_n$$

where

$$V_n = n^{-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2, \bar{X}_n = n^{-1} \sum_{i=1}^n X_i$$

$$\text{and } c_n = n^{-1} c(n + 1)^3 - (2n)^{-1} - (2n)^{-2}$$

First consider the case $\theta \neq \frac{1}{2}$.

We show that $h(N) \xrightarrow{D} N(0, 1)$ as $c \rightarrow 0$ where

$$h(N) = [N - c^{-\frac{1}{2}} \{\theta(1 - \theta)\}^{\frac{1}{2}}] / [c^{-\frac{1}{2}}(\theta - \frac{1}{2}) \{\theta(1 - \theta)\}^{-\frac{1}{2}}]$$

First note that $N/\lambda \rightarrow 1$ a.s. as $c \rightarrow 0$, where $\lambda = c^{-1/2} \{\theta(1-\theta)\}^{1/2}$ and that

$$n^{1/2} \phi^{-1} (V_n - \theta(1-\theta)) \xrightarrow{D} N(0,1) \text{ as } n \rightarrow \infty$$

where $\phi = 2(\theta - 1/2) \{\theta(1-\theta)\}^{1/2}$. From these and Anscombe (1952) follows

$$d(c)(V_N - \theta(1-\theta)) \xrightarrow{D} N(0,1) \text{ as } c \rightarrow 0,$$

where $d(c) = c^{-1/2} \{\theta(1-\theta)\}^{1/2} \phi^{-1}$. That Anscombe's continuity condition is satisfied, follows from the fact that $[(n-1)/n]V_n$ is a U-statistic, so that Corollary 4.2 of Geertsema (1970) implies

$$(2.1) \quad V_n = (n-1)^{-1} \sum_{i=1}^n (X_i - \theta)^2 + o(n^{-1} \log n) \text{ a.s.}$$

With $g(n) = n^{1/2} \phi^{-1}$, (1.4) becomes

$$(2.2) \quad m^{1/2} \phi^{-1} (m^{-1} \alpha(m+1)^3 - (2m)^{-1} - (2m)^{-2} - \theta(1-\theta)) = x$$

from which follows $m^{-1} \alpha(m+1)^3 \rightarrow \theta(1-\theta)$ as $c \rightarrow 0$, i.e. $m/\lambda \rightarrow 1$ as $c \rightarrow 0$.

Also from (2.2), as $c \rightarrow 0$

$$m^{-1} \alpha(m+1)^3 - \theta(1-\theta) = x m^{-1/2} \phi + o(1)$$

$$\text{i.e.} \quad c m^2 - \theta(1-\theta) = x m^{-1/2} \phi + o(1)$$

$$\text{i.e.} \quad (c^{1/2} m - \{\theta(1-\theta)\}^{1/2}) (c^{1/2} m + \{\theta(1-\theta)\}^{1/2}) = x m^{-1/2} \phi + o(1)$$

so that

$$c^{1/2} [m - c^{-1/2} \{\theta(1-\theta)\}^{1/2}] \rightarrow x \phi \{\theta(1-\theta)\}^{-1/2} 2^{-1} \{\theta(1-\theta)\}^{-1/2}$$

$$\text{i.e.} \quad h(m) \rightarrow x.$$

The left side of (1.5) is

$$c^{-1/2} \{\theta(1-\theta)\}^{1/2} \phi^{-1} [c N^2 - \theta(1-\theta)] - h(N) + o_p(1)$$

$$= 2^{-1} (\theta - 1/2)^{-1} \{\theta(1-\theta)\}^{-1/2} c^{1/2} [N - c^{-1/2} \{\theta(1-\theta)\}^{1/2}]^2 + o_p(1)$$

$$= o_p(1)$$

which can be seen as follows:

Using the definition of N and (2.1) we obtain

$$(N-2)c_{N-1} + (X_N - \theta)^2 + o(\log N) \leq \sum_{i=1}^N (X_i - \theta)^2 \\ \leq (N-1)c_N + o(\log N)$$

Since, for $1 \leq r < 2$, $N^{-1/r}[(N-1)c_N - (N-2)c_{N-1}] \rightarrow 0$ a.s. and

$$N^{-1/r}[\sum_{i=1}^N (X_i - \theta)^2 - N\theta(1-\theta)] \rightarrow 0 \text{ a.s. as } c \rightarrow 0, \text{ we have}$$

$$N^{-1/r}[(N-1)c_N - N\theta(1-\theta)] \rightarrow 0 \text{ a.s., which shows}$$

$$c^{1/2r}[N - c^{-1/2} \{\theta(1-\theta)\}^{1/2}] \rightarrow 0 \text{ a.s.}$$

Now choose $r = 4/3$. (See Gut (1974) for similar results).

In the case $\theta = 1/2$ our method yields the asymptotic distribution of N (Bhattacharya and Mallik (1973) give asymptotic bounds only). We show $h(N) \xrightarrow{D} -\chi_1^2$ as $c \rightarrow 0$, where $h(N) = 2[N - (4c)^{-1/2}] + 1$

Now $4n(V_n - 1/2) \xrightarrow{D} -\chi_1^2$ as $n \rightarrow \infty$, $g(n) = 4n$, $\lambda = (4c)^{-1/2}$, $d(c) = 2c^{-1/2}$ and (1.4) becomes

$$(2.3) \quad 4m[m^{-1}\alpha(m+1)^3 - (2m)^{-1} - (2m)^{-2} - 1/4] = x$$

from which follows $m/\lambda \rightarrow 1$ as $c \rightarrow 0$.

Note that $c_m - cm^2 = 3cm - (2m)^{-1} + f(c)$ where $f(c) = o(c^\delta)$ and $0 < \delta < 1$. Hence (2.3) may be written

$$cm^2 - 1/4 = (4m)^{-1}x - 3cm + (2m)^{-1} - f(c)$$

so that $h(m) \rightarrow x$ as $c \rightarrow 0$.

The left side of (1.5) may be written as

$$(2.4) \quad 2c^{1/2}(N - (4c)^{-1/2})^2 + o_p(1) = o_p(1)$$

which is proved as follows:

Using the identity $V_n = n^{-1}(n-1)V_{n-1} + R_n$

where $R_n = n^{-2}(n-1)(\bar{X}_{n-1} - X_n)^2$ we get

$$(2.5) \quad N^{-1}(N-1)c_{N-1} + R_N < V_N \leq c_N$$

From (2.1) follows

$$N(\log N)^{-1}(V_N - \frac{1}{2}) \rightarrow 0 \text{ a.s. as } c \rightarrow 0.$$

$$\text{Also: } N(\log N)^{-1}[c_N - N^{-1}(N-1)c_{N-1} - R_N] \rightarrow 0 \text{ a.s.}$$

Hence, from (2.5),

$$(\log N)^{-1}(N-1)c_{N-1} - N(\log N)^{-1}\frac{1}{2} \rightarrow 0 \text{ a.s.}$$

so that

$$N(\log N)^{-1}(c^{\frac{1}{2}}N - \frac{1}{2}) \rightarrow 0 \text{ a.s.}$$

This implies (2.4).

Remark:

The stopping rule considered in Theorem 2 of Bhattacharya and Mallik (1973) is defined as follows: Let N be the first n such that $V_n \leq c(n-1)n$. It is clear that in a way analogous to the case $\theta \neq \frac{1}{2}$ above the asymptotic distribution of N can be derived. It is worth nothing that this approach does not require the assumption of normality. Instead, a representation such as (2.1), which requires only a finite fourth moment, is used.

- (3) The following stopping rule was considered by Siegmund (1968): Let $X_1, X_2, \dots$ be i.i.d. random variables with common expectation $\mu > 0$ and common variance $\sigma^2 < \infty$, $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$ and define N the first n such that

$$\bar{X}_n \geq c^{-1}n^{-(1-\alpha)}, 0 \leq \alpha < 1.$$

Let $h(N) = (N - \lambda)/[(1 - \alpha)^{-1}\lambda^{\frac{1}{2}}\mu^{-1}\sigma]$, where $\lambda = (\mu c)^{-1}/(1-\alpha)$. We show how Siegmund's result, $h(N) \xrightarrow{D} N(0,1)$, may be obtained, using our methods.

First note that $N/\lambda \rightarrow 1$ a.s. as $c \rightarrow 0$, that $n^{\frac{1}{2}}\sigma^{-1}(\bar{X}_n - \mu) \xrightarrow{D} N(0,1)$ as $n \rightarrow \infty$ and $\lambda^{\frac{1}{2}}\sigma^{-1}(\bar{X}_N - \mu) \xrightarrow{D} N(0,1)$ as $c \rightarrow 0$. Here $g(n) = n^{\frac{1}{2}}\sigma^{-1}$, $d(c) = \lambda^{\frac{1}{2}}\sigma^{-1}$. Then (1.3) follows as in Siegmund (1968).

The left side of (1.5) becomes

$$\begin{aligned} & \lambda^{\frac{1}{2}} \sigma^{-1} \mu[(\lambda/N)^{1-\alpha} - 1] - h(N) \\ = & \lambda^{\frac{1}{2}} \sigma^{-1} \mu(1 - \alpha)(1 - N/\lambda)^2 [\lambda/N - \frac{1}{2}(\lambda/N)^2 \alpha(1 + \theta \lambda/N)^{-(1+\alpha)}] \end{aligned}$$

using a Taylor expansion of $(\lambda/N)^{1-\alpha} - 1$ about the point 1, where $|\theta| \leq 1$.

It is now sufficient to prove that

$$\lambda^{\frac{1}{2}}(1 - N/\lambda) = o_p(1)$$

Using the definition of N we see, as before,

$$\mu \lambda^{(r-1)/r} (N/\lambda)^\alpha [1 - (N/\lambda)^{1-\alpha}] \rightarrow 0 \text{ a.s. as } c \rightarrow 0.$$

Expanding $1 - (N/\lambda)^{1-\alpha}$ in a Taylor series, we obtain

$$\lambda^{(r-1)/r} (1 - N/\lambda) \rightarrow 0 \text{ a.s. as } c \rightarrow 0.$$

Now choose $r = 4/3$.

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